approach_1, approach_2 and approach_4 trained with the annealed self-logit bias on OpenWebText: structure, initialization, training and failure modes

Experiment log, branch `attention-experiments`

2026-10-20

Summary

On FineWeb-Edu, approach_1, approach_2 and approach_4 all collapsed to the unigram loss. The 2026-10-19 report showed that three_residual, which collapsed the same way, trains if a bias b is added to every head's self logit at initialization and lowered linearly from 8 to 0 over the first 1000 steps. This report applies the same recipe, unchanged, to the three collapsed variants on OpenWebText (2.5B tokens, Chinchilla budget).

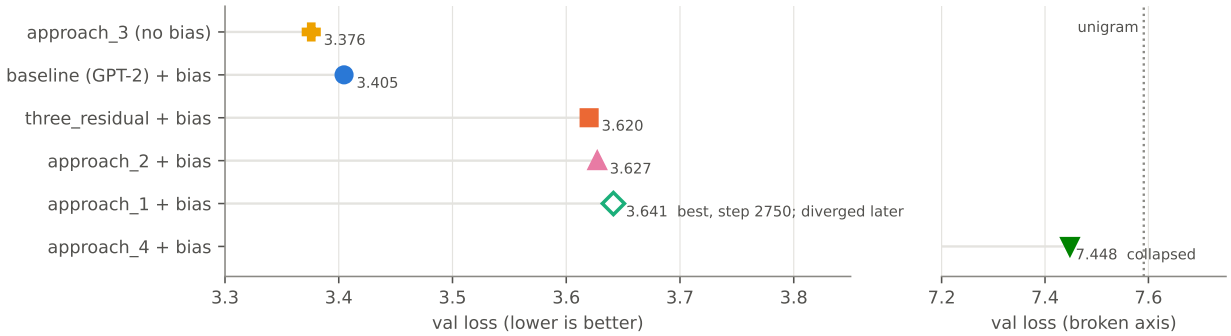


Figure 1: Validation loss after 2.5B OpenWebText tokens on the full 20M-token split. approach_1 (open diamond) did not finish: its value is the best 4M-token evaluation, at step 2750, before it diverged. The right panel is a broken axis for the collapsed approach_4; the dotted line is the unigram model.

- **approach_2: rescued.** It trains stably to 3.627, level with three_residual (3.620) and 0.223 nats behind GPT-2 trained with the same bias (3.405). Without the bias it stays at the unigram level (7.65 at step 500).
- **approach_1: rescued, then unstable.** It learned fastest of the new variants: at step 2500 it was only 0.10 nats behind the control. From step ∼2750 its gradient norm grew from 0.74 to 18.1 (medians), and around step 3600 the loss blew up. Best value 3.64 at step 2750; the run was stopped at step 3690.

- **approach_4: not rescued.** It trained well while the bias was on (4.89 at step 750, ahead of approach_2), became unstable as the bias fell below about 2, and collapsed within 250 steps of the bias reaching 0 (7.67 at step 1250). It ended at 7.448, near the unigram loss.
- **Why approach_4 fails is not that it cannot learn the right attention.** Checkpoints show that its own query-key logits learn to attend to the token itself as fast as approach_2's do (mean self-weight with the bias removed: 0.64 vs. 0.67 at step 750). It then loses this attention in one burst after step 1000, while its gradient norm is far above the clipping threshold.
- **The surviving models attend mostly to themselves.** With the bias at 0, approach_2 puts a mean weight of 0.83 on the token itself (122 of 144 heads above 0.5); three_residual: 0.61; GPT-2: 0.16.

1 Setup

Data	OpenWebText, GPT-2 BPE; train 2.69B tokens, validation 20M tokens (same split as the 2026-10-19 report)
Budget	2.5B tokens: 4768 steps $\times$ 524,288 tokens
Optimizer	AdamW, $\beta = (0.9, 0.95)$, weight decay 0.1 (matrices only), gradient clip 1.0
Learning rate	linear warmup over 250 steps to 6×10^{-4}, cosine decay to 6×10^{-5}
Batch	16 micro-batches of 32×1024 tokens; bf16 autocast; `torch.compile`; one H100 per run
Bias schedule	$b(t) = 8 \max(0, 1 - t/1000)$ on every head's self logit, a non-learned buffer
Seed and order	seed 1337 and the same sequential data order in every run
Evaluation	4M validation tokens every 250 steps; the full 20M tokens at the end

All numbers are on the same OpenWebText validation split and are comparable with the 2026-10-19 report (the three_residual, baseline and approach_3 runs shown here are those runs). They are not comparable with the FineWeb-Edu numbers of the 2026-10-15/16 reports. The unigram model scores 7.591.

Runs in this report.

Run	Purpose	Directory under `runs/openwebtext/`
approach_1, approach_2, approach_4 + bias	main runs, 2.5B tokens	`approach_{1,2,4}_diag8_anneal1000`
approach_1, approach_2, approach_4, no bias	collapse reference, 500 steps	`approach_{1,2,4}_nobias_s500`
approach_2, approach_4 + bias, repeated	checkpoints at steps 500–1250	`approach_{2,4}_diag8_repro`
baseline + bias, three_residual + bias, approach_3	references (2026-10-19)	`baseline_diag8_anneal1000`, `tr_diag8_anneal1000`, `approach_3`

The main runs keep only their latest checkpoint, so the checkpoints around the end of annealing come from the repeated runs: same seed, data order and settings, stopped at step 1250. They reproduce the main runs closely, though not bit for bit (approach_2 at step 250: 6.12 vs. 6.08; approach_4 at step 1000: 5.16 vs. 5.06), and the repeated approach_4 collapses at the same point.

2 Structure

2.1 The blocks

For head h (12 heads, $d = 64$), X_h is the head's d-dimensional slice of the raw residual stream X (before LN_1), and Q_h, K_h, V_h are computed from $\mathrm{LN}_1(X)$ as in GPT-2. During pretraining the causal attention matrix is

$$A_h = \mathrm{softmax}\Big(\tfrac{1}{\sqrt{d}}\, Q_h K_h^\top + b\, I + \mathrm{mask}\Big),$$

and from step 1000 on $b = 0$, so every model is exactly as defined below. Only the attention sublayer differs; the MLP sublayer is standard in all variants, $X^{(\ell+1)} = X' + \mathrm{MLP}(\mathrm{LN}_2(X'))$.

$$\text{baseline (GPT-2):} \quad X' = X + \mathrm{Concat}_h[A_h V_h]\, W_O + b_O \tag{1}$$

$$\text{approach_1:} \quad Y = \mathrm{Concat}_h[A_h(V_h + X_h)], \qquad X' = YW_O + b_O + Y \tag{2}$$

$$\text{approach_2:} \quad Y = \mathrm{LN}\big(\mathrm{Concat}_h[A_h(V_h + X_h)]\big), \qquad X' = YW_O + b_O + Y \tag{3}$$

$$\text{approach_4:} \quad X' = \mathrm{Concat}_h[A_h V_h]\, W_O + b_O + \mathrm{Concat}_h[A_h X_h] \tag{4}$$

The LayerNorm in approach_2 has no affine parameters. All three new variants have exactly GPT-2's parameters (124,475,904): they reuse W_Q, W_K, W_V and W_O, and only re-route X. For reference, three_residual is $Z_h = (I + A_h)X_h$, $Y = \mathrm{Concat}_h[Z_h(I + W_{vh})]$, $X' = Y(I + W_O) + b_O$, and approach_3 is $X' = X + \mathrm{Concat}_h[A_h(V_h + X_h)]W_O + b_O$.

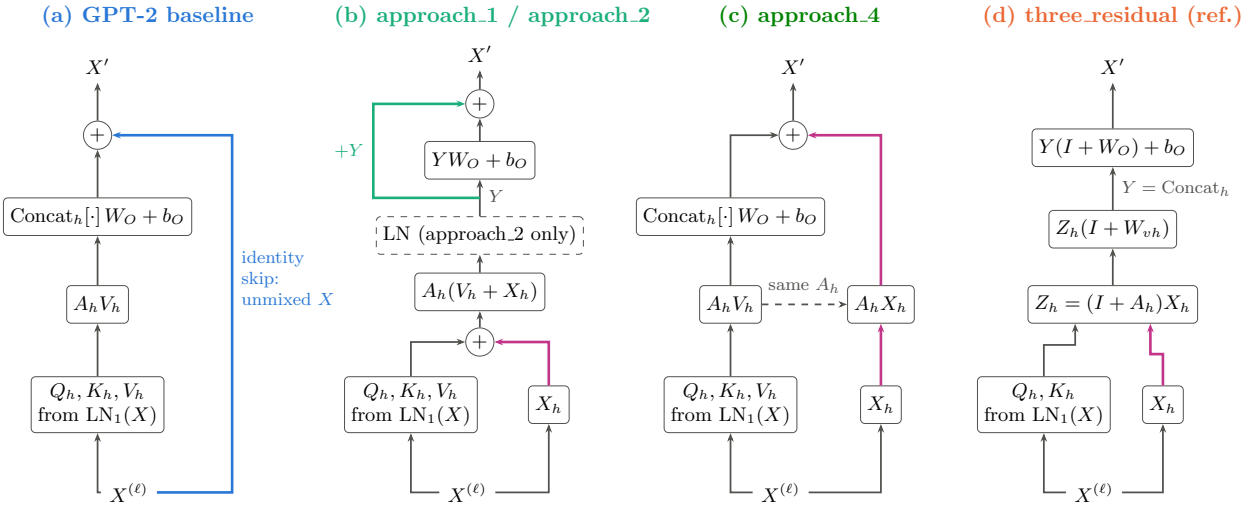


Figure 2: Attention sublayers. Blue: GPT-2's identity skip, which carries each token's state to the next layer unmixed. Magenta: the raw stream X_h entering the attention average A_h at full scale. Green: the skip around W_O in approach_1/approach_2, which carries Y, itself already averaged by A_h. In approach_1, approach_2 and approach_4, every path from $X^{(\ell)}$ to X' passes through some A_h; none has an unmixed copy of X. In three_residual (d) the identity inside $(I + A_h)$ does carry X_h unmixed, but at the same scale as $A_h X_h$.

2.2 Expanding the outputs

Dropping the biases and writing each output as a sum of terms shows what reaches the next layer:

Variant	Attention sublayer output X'	Unmixed X?
baseline	$X + \mathrm{Concat}_h[A_h V_h]W_O$	yes
approach_3	$X + \mathrm{Concat}_h[A_h V_h]W_O + \mathrm{Concat}_h[A_h X_h]W_O$	yes
three_residual	$\mathrm{Concat}_h\big[(X_h + A_h X_h)(I + W_{vh})\big](I + W_O)$	yes*
approach_1	$\mathrm{Concat}_h[A_h X_h] + \mathrm{Concat}_h[A_h V_h] + \mathrm{Concat}_h[A_h X_h]W_O + \mathrm{Concat}_h[A_h V_h]W_O$	no
approach_2	$\mathrm{LN}\big(\mathrm{Concat}_h[A_h X_h + A_h V_h]\big)\,(I + W_O)$	no
approach_4	$\mathrm{Concat}_h[A_h X_h] + \mathrm{Concat}_h[A_h V_h]W_O$	no

* at the same scale as the averaged term $A_h X_h$

- **At initialization the terms differ a lot in size.** V_h comes from $\mathrm{LN}_1(X)$, which has unit scale, while the token and position embeddings have standard deviation 0.02; and W_O shrinks whatever passes through it by about 10×. Measured norms at layer 0 (4 validation sequences, identical for the three variants):

	X	$\mathrm{Concat}_h[A_h X_h]$	$\mathrm{Concat}_h[A_h V_h]$	$\mathrm{Concat}_h[A_h X_h]W_O$	$\mathrm{Concat}_h[A_h V_h]W_O$
no bias	49	5	90	1	10
$b = 8$	49	41	801	5	91

  So the variants differ in how fresh content enters the stream: approach_1 and approach_2 add the averaged values $\mathrm{Concat}_h[A_h V_h]$ at full scale in every layer, approach_4 only through W_O. By layer 11 (with $b = 8$) the stream has grown to a norm of 1785 in approach_1 and 1732 in approach_2, but only 388 in approach_4.

- **Every term is an average over the prefix.** With near-uniform attention all of them are close to the prefix mean (the random values largely cancel, which is why $\mathrm{Concat}_h[A_h V_h]$ is 90 rather than 801), and 12 layers of this make every position alike (Section 3). With $b = 8$, $A_h \approx I$, so $A_h X_h \approx X_h$ and $A_h V_h \approx V_h$: the bias temporarily turns the averages into per-token terms and stands in for the missing identity skip.

- **approach_1 contains approach_4.** approach_1 has both of approach_4's terms and adds two more, $\mathrm{Concat}_h[A_h V_h]$ (the averaged values at full scale, with no W_O) and $\mathrm{Concat}_h[A_h X_h]W_O$. So the two variants differ only by extra terms; neither has a path the other lacks around the attention.

- **approach_2** is approach_1 with the attention output normalized before W_O and before the skip, so the scale of Y is fixed regardless of the scale of X.

The 2026-10-16 rule ("a variant trains only if each token's state has an unmixed path to the next layer") classified all three as untrainable. The results below show that this rule describes initialization, not the architecture: approach_1 and approach_2 train with no unmixed path once they start from a good point.

3 Initialization

The probe runs the freshly initialized model on 8 validation sequences and records, after every layer, the residual stream's *distance from rank 1*, $\|X - \bar{X}\|/\|X\|$, where $\bar{X}$ repeats the mean over positions (positions below 64 skipped). In every earlier case a layer-11 value of about 0.15 or less predicted collapse.

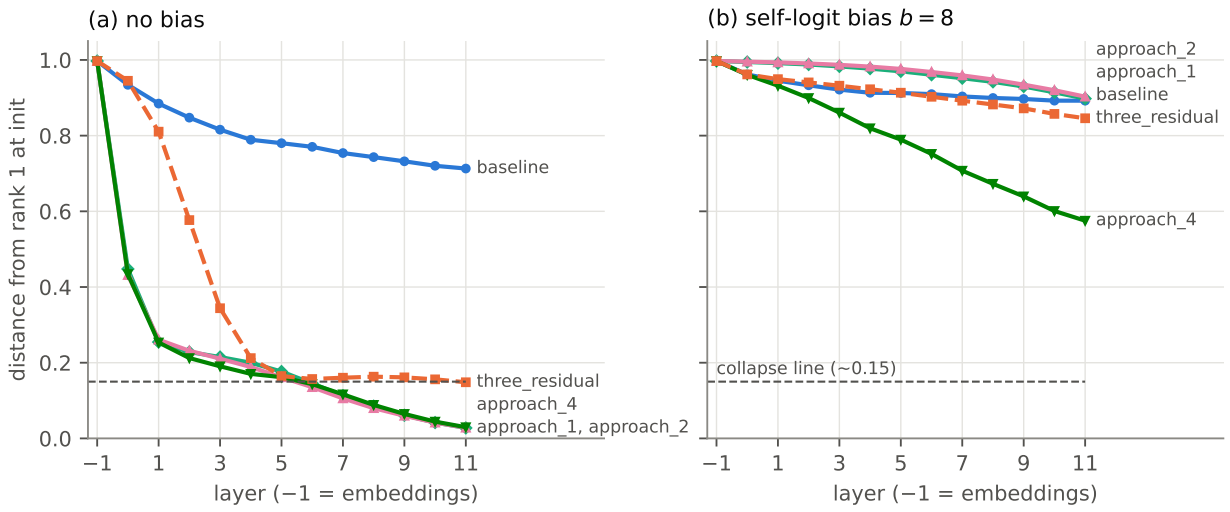


Figure 3: Distance from rank 1 at initialization. (a) Without the bias, approach_1, approach_2 and approach_4 lose almost all position-specific information by layer 5 and end at about 0.03, below even three_residual. (b) With $b = 8$ all variants stay far above the collapse line. approach_4 decays fastest (0.575 at layer 11), but it is still well clear of the line.

Layer-11 distance from rank 1 at init	no bias	$b = 8$	val loss at step 500, no bias
baseline	0.713	0.892	—
approach_1	0.028	0.898	7.65
approach_2	0.028	0.903	7.65
approach_4	0.030	0.575	7.56

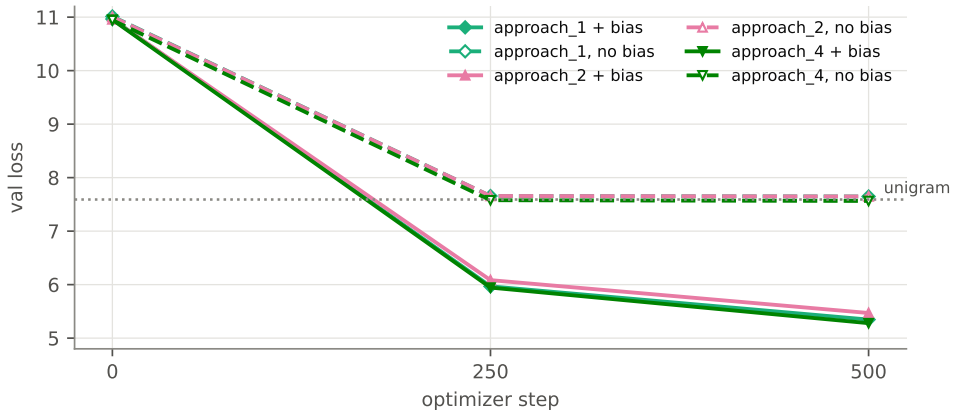


Figure 4: First 500 steps with and without the bias. Without it (dashed, open markers), all three variants flatten at the unigram loss within 250 steps on OpenWebText too, confirming the FineWeb-Edu result on this data. With it (solid) they learn like the healthy models. (approach_1 + bias is hidden under approach_4 + bias.)

4 Pretraining

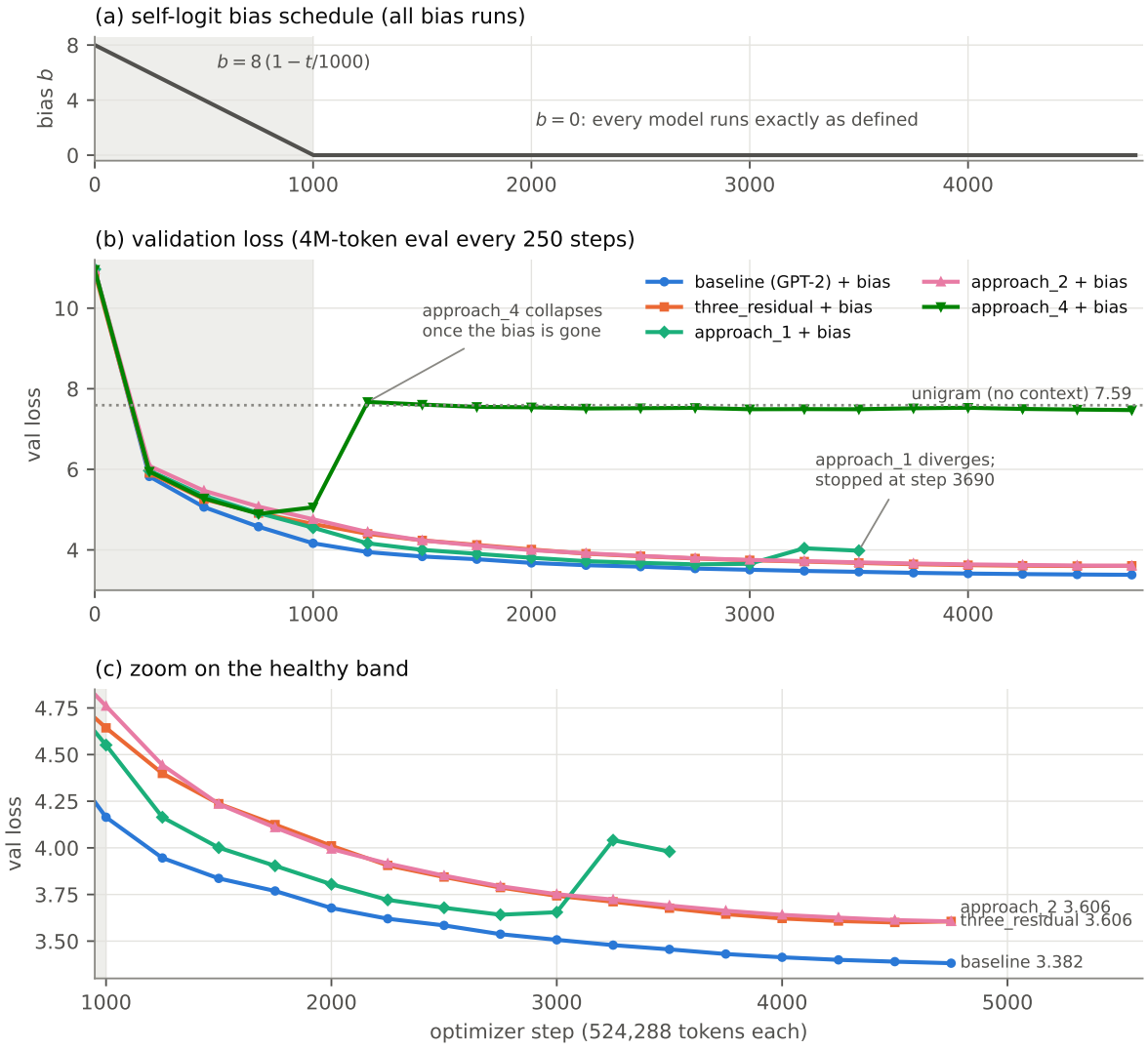


Figure 5: (a) Bias schedule; the shaded band marks steps 0–1000, while the bias is non-zero. (b) Validation loss of all bias runs. (c) Zoom on the healthy band, steps 1000–4750; labels give the last 4M-token value (the final 20M-token values are in Figure 1).

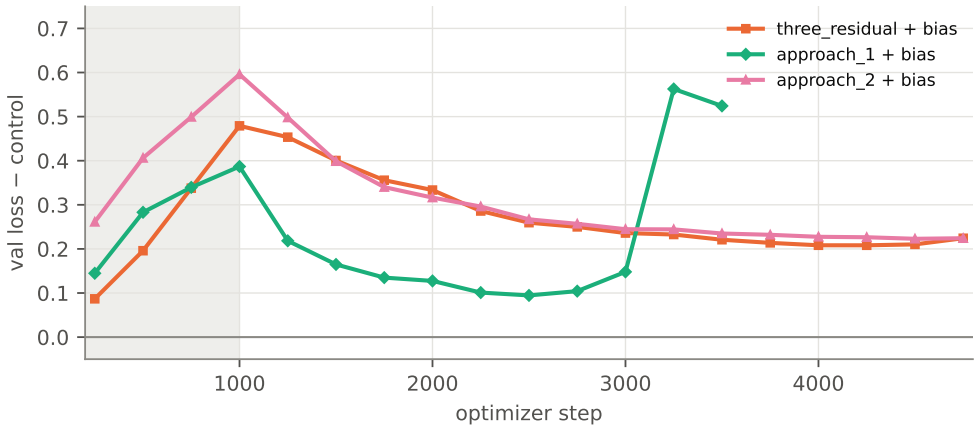


Figure 6: Validation loss minus the GPT-2 control at the same step. All three healthy variants lag most at step 1000, when the bias reaches 0, and then close in. approach_1 closes fastest, to 0.10 nats at step 2500, before it diverges. approach_2 and three_residual follow almost the same curve and both settle near 0.22.

Val loss	step 750 ($b = 2$)	step 1000 ($b = 0$)	step 1250	step 2500	final (20M)
baseline + bias	4.58	4.16	3.95	3.58	3.405
three_residual + bias	4.91	4.64	4.40	3.84	3.620
approach_1 + bias	4.92	4.55	4.16	3.68	diverged (best 3.64)
approach_2 + bias	5.07	4.76	4.44	3.85	3.627
approach_4 + bias	4.89	5.06	7.67	7.52	7.448

Throughput. The bias needs an additive attention mask, which rules out PyTorch's fastest attention kernel. approach_2 trained at 219k tokens/s. approach_4 needs two attention calls when the bias is on (Section 7) and ran at 166k tokens/s.

5 Stability

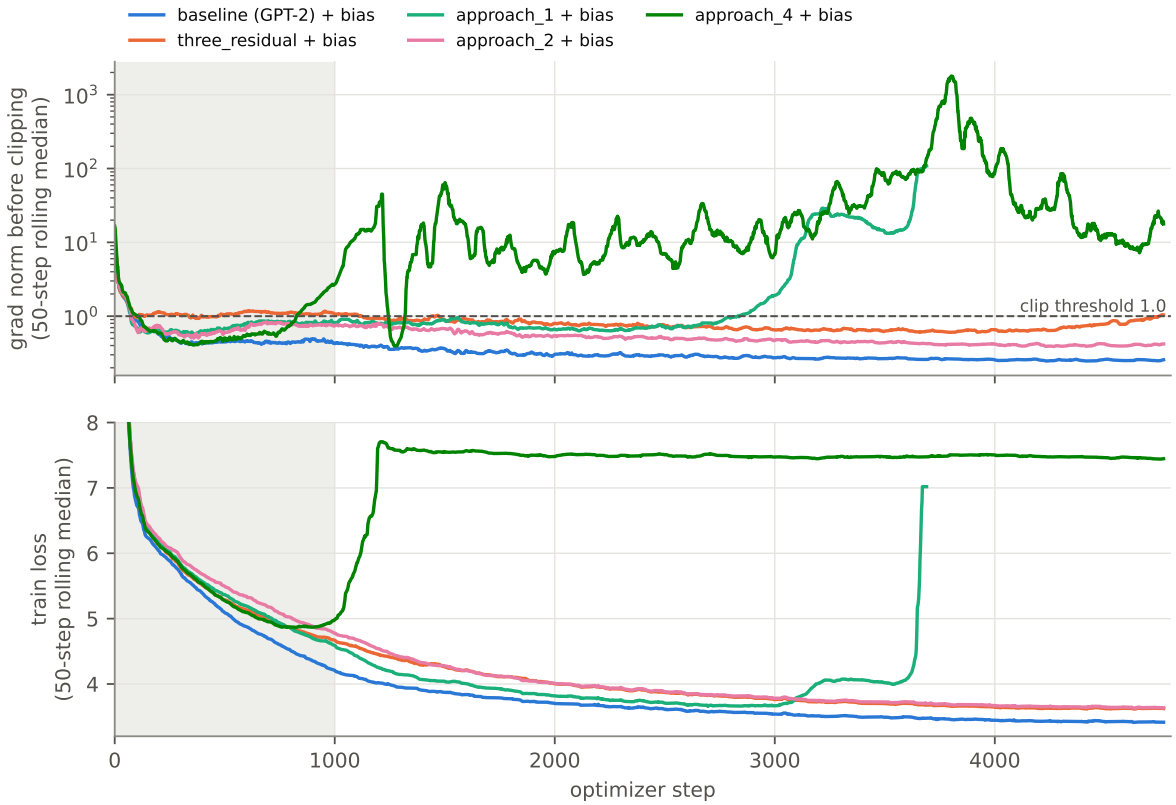


Figure 7: Gradient norm before clipping (top, log scale) and training loss (bottom), both as 50-step rolling medians. The control, three_residual and approach_2 stay below the clip threshold for the whole run. approach_4 rises above it from step ∼800 and approach_1 from step ∼2900; each loss blows up after its gradient norm has risen by about an order of magnitude.

Median gradient norm (pre-clip)	steps 3000–3500
baseline + bias	0.27
three_residual + bias	0.65
approach_2 + bias	0.45
approach_1 + bias	18.1 (steps 1000–2500: 0.74)

5.1 approach_4 at the end of annealing

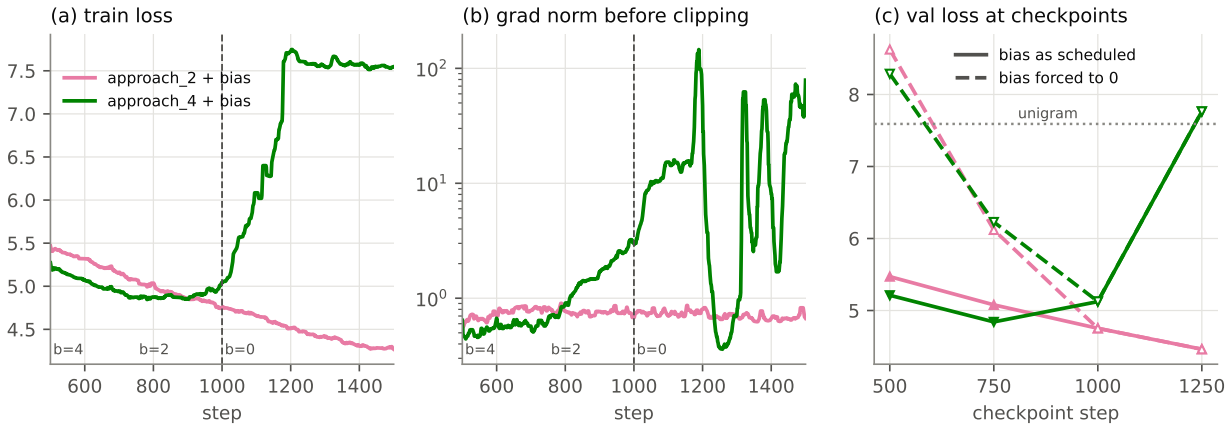


Figure 8: Steps 500–1500 for approach_2 and approach_4. (a) Training loss and (b) gradient norm, 20-step rolling medians; the bias values are marked at the bottom. (c) Validation loss (2M tokens) of the checkpoints from the repeated runs, with the bias at its scheduled value (solid) and forced to 0 (dashed). Where the two lines meet, the model no longer needs the bias.

	approach_2			approach_4		
Checkpoint	val, bias on	val, $b = 0$	self-weight, $b = 0$	val, bias on	val, $b = 0$	self-weight, $b = 0$
step 750 ($b = 2$)	5.08	6.12	0.67	4.84	6.23	0.64
step 1000 ($b \approx 0$)	4.75	4.75	0.84	5.12	5.13	0.72
step 1250 ($b = 0$)	4.46	4.46	0.85	7.76	7.76	0.06

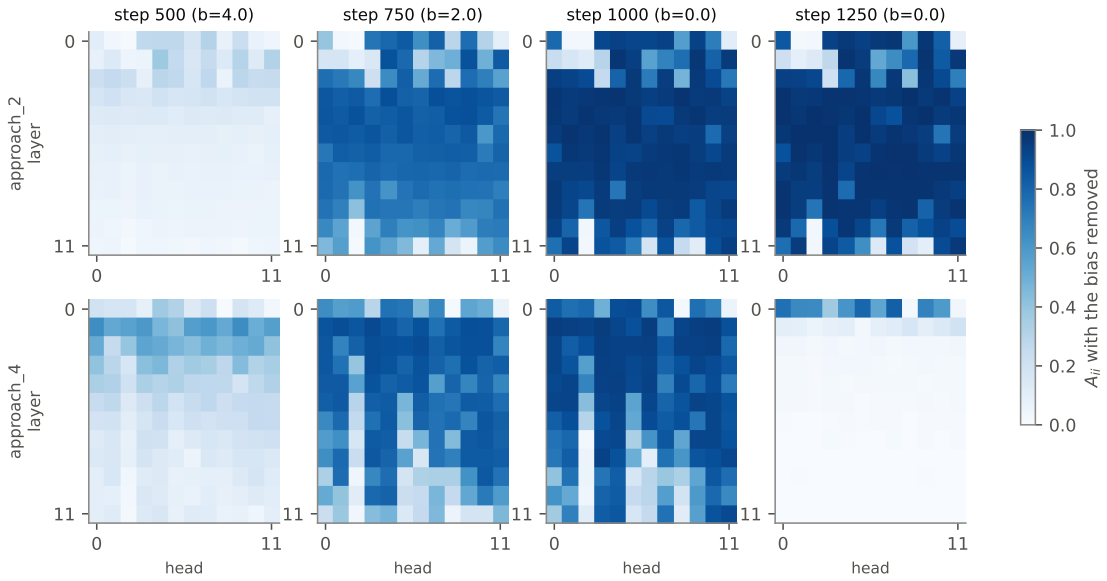


Figure 9: Self-weight A_{ii} of every head (rows: layers 0–11, columns: heads) computed from the learned query-key logits alone, i.e. with the bias removed. Both variants learn self-attention at about the same pace between steps 500 and 1000. approach_4's deep layers learn less of it (layers 6–11: 0.62 at step 1000 vs. 0.86 for approach_2). Between steps 1000 and 1250 approach_4 loses almost all of it.

What the checkpoints show:

- **Both variants take over the bias's job.** At step 500 neither works without the bias (validation loss above the unigram level when it is removed). By step 1000 both have learned query-key logits that put most attention on the token itself, and removing the remaining bias changes nothing.
- **approach_4 starts degrading before the bias is gone.** Its gradient norm leaves the clip threshold at about step 800 ($b \approx 1.6$; median 0.60 at steps 650–700 vs. 8.0 at 1000–1050), and its loss is higher at step 1000 than at step 750, although it no longer depends on the bias at step 1000.
- **The collapse is a loss of the learned self-attention.** Between steps 1000 and 1250 the self-weight falls from 0.72 to 0.06 (heads above 0.5: 119 $\rightarrow$ 9), and the residual stream becomes rank 1 (layer-11 distance 0.004). The model ends where the unbiased model starts, and it never recovers: its attention afterwards is sharp but shared across queries, the pattern seen in the 2026-10-16 approach_4 run.

So the bias fixes approach_4's starting point, as it does for the other variants, but approach_4 cannot hold the state it reaches. Why it is unstable and approach_2 is not has not been established. Candidates: approach_2's LayerNorm fixes the scale of Y; approach_1 and approach_2 add fresh, full-scale per-token content $\mathrm{Concat}_h[A_h V_h]$ in every layer, while approach_4's update is almost a pure average of the existing stream X (its values enter only through W_O), so an error in A_h is not diluted by anything new.

5.2 approach_1's late divergence

approach_1 was healthy for 2000 steps after the bias reached 0, and better than approach_2 and three_residual. Its gradient norm started rising at about step 2750, crossed the clip threshold near step 2900 and reached medians of 18.1 after step 3000, with single steps above 300. Validation loss went from 3.65 at step 3000 to 4.04 at step 3250, and the training loss climbed above 5 by step 3630. The step-3000 checkpoint is kept (`ckpt_step3000.pt`). Unlike approach_2, approach_1 has no normalization on Y, and Y feeds the stream at full scale through the skip around W_O; a growing stream scale is the first thing to check. A rerun with a lower peak learning rate would show whether this is a learning-rate problem or a property of the block.

6 The trained models

All numbers in this section are averages over 8 validation sequences and query positions ≥ 64; heatmaps show the first 48 tokens of one sequence. The bias is 0 in every model shown.

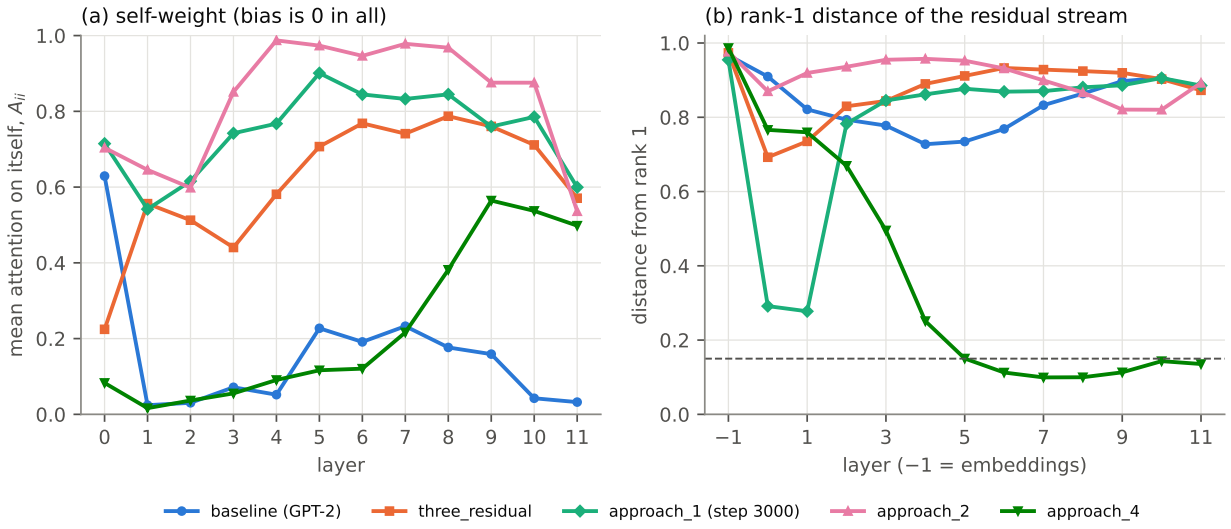


Figure 10: (a) Mean attention on the token itself, per layer. (b) Distance from rank 1 of the residual stream. approach_1 is shown at step 3000, its last checkpoint before the divergence.

	mean A_{ii}	heads with $A_{ii} > 0.5$	layer-11 distance from rank 1
baseline + bias	0.16	14 / 144	0.879
three_residual + bias	0.61	94 / 144	0.873
approach_1 + bias (step 3000)	0.75	112 / 144	0.885
approach_2 + bias	0.83	122 / 144	0.893
approach_4 + bias (collapsed)	0.23	30 / 144	0.136

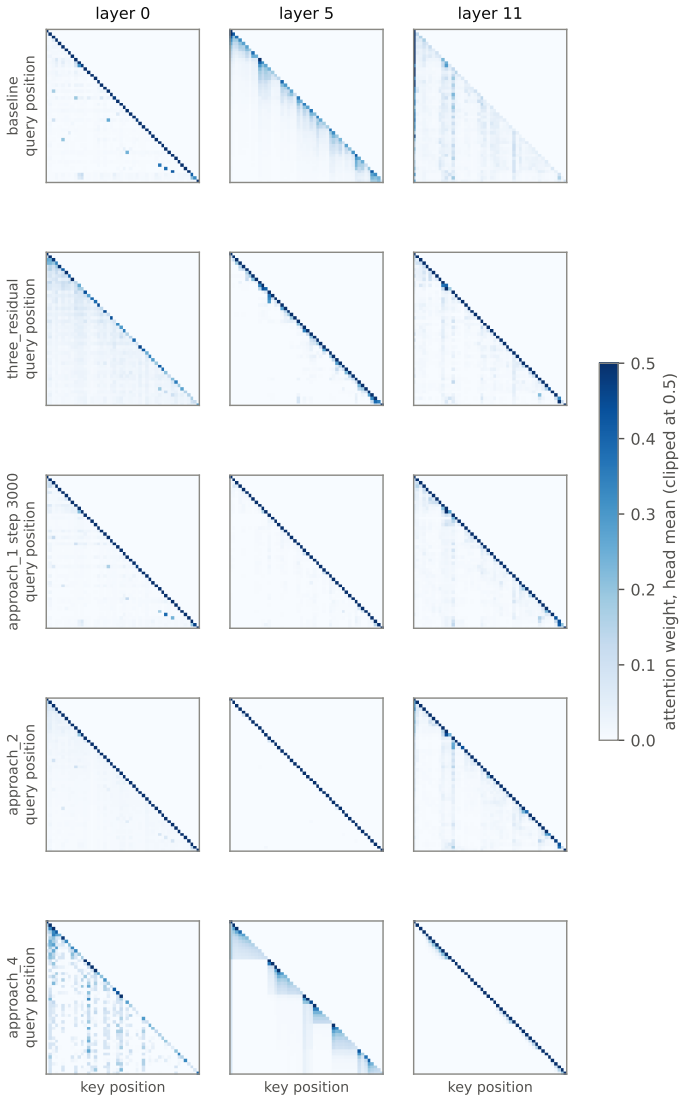


Figure 11: Attention averaged over heads at layers 0, 5 and 11. approach_1 and approach_2 are almost pure self-attention at every depth, with a few weak stripes on earlier tokens. The GPT-2 control shows the usual patterns (previous tokens, a column on the first token). The collapsed approach_4 has block-shaped attention in layer 5: queries in a block all read the same position.

- **The surviving variants keep each token's state by attending to itself.** approach_2 puts 0.95–0.99 on the token itself in layers 4–8. This is the same strategy three_residual learned, carried further: without an identity skip, $A_h \approx I$ is the only way to keep a token's own state dominant.
- **Little attention is left for moving information.** Only 22 of approach_2's 144 heads put less than half their weight on the token itself (control: 130). That approach_2 still reaches 3.627 means a small amount of off-diagonal attention, plus the MLPs, does most of the work. This is also a plausible reason for the remaining 0.22-nat gap, but it has not been tested.
- **approach_1's stream briefly narrows.** At step 3000 its distance from rank 1 drops to 0.29 after layers 0 and 1 and recovers by layer 2 (Figure 10b). No other healthy model shows

this.

7 Code changes

- `train/gpt2_torch.py`: the self-logit bias now also applies to approach_4 and approach_5 (it was blocked by an assert). Under `torch.compile`, the original single attention call on the concatenated $[V_h \mid X_h]$ (value width 128, query width 64) with an additive mask gave NaN gradients at step 0, while eager mode was fine. When the bias is on, $A_h V_h$ and $A_h X_h$ are now computed with two separate attention calls; the compiled run then matches eager mode exactly. Without the bias the original single call is kept.
- `train/train.py`: `--save_steps` keeps extra model-only checkpoints at given steps.
- `train/annealed_approaches_analysis.py` (measurements), `train/make_annealed_approaches_report.py` (figures, `numbers.tex`); copies are in `code/`.

Main run command (one GPU each):

```
python train.py --variant approach_2 --diag_bias 8 --anneal_steps 1000
                --data <openwebtext dir> --out <run dir>
```

8 Limitations

- One seed per configuration. The repeated runs differ from the main runs by up to 0.1 nats around step 1000, so differences of that size between variants early in training are not reliable.
- approach_1's best value is a 4M-token evaluation; the other final values use the full 20M tokens. On the runs that finished, the two differ by about 0.02.
- The annealing checkpoints come from repeated runs, not from the main runs.
- Only one bias schedule ($b = 8$, linear over 1000 steps) was tried for the new variants.
- The explanations for approach_4's and approach_1's instabilities are untested hypotheses.

9 Next steps

1. **approach_1 with a lower peak learning rate** (3×10^{-4}), to see whether the late divergence is a learning-rate problem. Track the residual-stream norm per layer during training.
2. **approach_4 with a slower anneal** (e.g. over 3000 steps), or a small floor (b stays at 1–2), to test whether it can hold its self-attention at all without the bias.
3. **Head-capacity test** for approach_2 and three_residual: give the block a cheap way to keep the token's state (for example a learned scalar on an identity term), and see whether the heads then move more information and the 0.22-nat gap shrinks.
4. A plain GPT-2 baseline on OpenWebText, for a reference without the bias schedule.